MOHAMMAD AREEB QAZI

Research Engineer | Video Generation, World Models & Large-Scale ML Systems
Abu Dhabi, United Arab Emirates
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Education

Mohamed Bin Zayed University of Artificial Intelligence — **August 2022 – May 2024**
Master of Science (Computer Vision) — Fully Funded Scholarship — *GPA: 3.67/4*

Aligarh Muslim University — **August 2019 – May 2022**
Bachelor of Science (Computer Science) — *CPI: 8.72/10*

Skills

ML & Deep Learning: PyTorch, Transformers, Diffusers, DiT, Video Diffusion Models, LoRA/PEFT, Model Merging, Continual Learning, Computer Vision, LLMs
Data Engineering & Pipelines: Large-Scale Data Collection, Filtering & Quality Validation, Dataset Curation, Preprocessing at Scale, Distributed Data Processing
Training & Evaluation: Distributed Training (FSDP, DeepSpeed), GPU Cluster Management, Training Pipeline Debugging & Profiling, Model/Dataset Evaluation
Edge & Deployment: Model Quantization, Pruning, Knowledge Distillation, Inference Optimization
Programming & Tools: Python, C++, C, Git, Docker, VSCode, LaTeX
AI Agents & RAG: LangChain, CrewAI, RAG Pipelines, LLM Orchestration
Languages: English (IELTS 7.5), Hindi, Urdu

Research Interests

World Models, Video Generation, Diffusion Transformers (DiT), Multimodal Foundation Models, Large-Scale Data Curation & Filtering, Continual Learning, Model Merging, Efficient Fine-tuning

Professional Experience

The Institute of Foundation Models, UAE — **November 2025 – Present**
Research Engineer — *Abu Dhabi, UAE*

- Conducted architectural research on video diffusion models (LTX-Video, LTX-2) — inference pipelines, Video VAE design, causal convolutions, RoPE-based 3D positional encoding — to inform pre-training data and pipeline requirements for **PAN V2 World Model**.
- Designed and built the end-to-end pre-training data pipeline for PAN V2: large-scale multimodal video ingestion, cleaning, filtering, and transformation across distributed infrastructure.
- Engineered scalable preprocessing workflows — frame extraction, temporal sampling, bucketed resolution/frame-rate batching, metadata alignment, and multimodal synchronization — and optimized distributed data processing across GPU clusters to cut preprocessing bottlenecks and raise training throughput.
- Defined dataset curation and quality-validation criteria (temporal coherence, diversity, robustness) to filter and evaluate training data for self-supervised world model learning.
- Supported large-scale training runs: debugged data pipeline issues, profiled performance, and improved compute utilization; built interactive demos for **PAN V1 World Model** and **PAN Game Model** for stakeholder evaluation.

Emirates Post (7X), UAE — **August 2024 – October 2025**
AI Engineer — *Dubai, UAE*

- Designed an address-parsing data pipeline using a RAG framework, converting raw UAE address inputs (including Arabic script) into structured, geocoded data at production scale.
- Led design and development of a modular, multi-agent AI system using open-source LLMs; engineered specialized agents (Tracking, Finding Branch, Knowledge) to manage complex real-world logistics tasks.
- Conducted R&D for LLM-based email automation and chatbot solutions, improving real-time responsiveness for the WAYN App.

Mohamed Bin Zayed University of Artificial Intelligence, UAE — **May 2024 – August 2024**
Research Associate — *Abu Dhabi, UAE*

- Developed continual learning strategies for Vision-Language models using a mixture-of-experts approach to dynamically fine-tune pre-trained networks while reducing computational overhead.
- Designed a novel framework to mitigate catastrophic forgetting, significantly enhancing model adaptability across sequential tasks.

Mohamed Bin Zayed University of Artificial Intelligence, UAE 🔗 **August 2022 – May 2024**
Student Researcher (BioMedAI Lab) *Abu Dhabi, UAE*

- Designed **DynaMMo** (class-incremental learning via lightweight task adapters, merged post-hoc) and **MedMerge** (kernel-level model merging with automated merge-weight search), two pipelines for consolidating and evaluating models trained on sequential medical imaging tasks.
- Co-created **XReal**, a controllable diffusion workflow for synthetic chest X-ray generation, and **SDICE**, a diversity index for evaluating synthetic medical datasets against real cohorts using contrastive encoders.
- Led a practical survey and benchmark of continual learning in medical imaging; standardized data protocols, reproducibility scripts, and evaluation best practices.
- Developed a multi-task ultrasound model (U-Net + classification/regression heads) for joint ROI detection and fetal biometric estimation, achieving MAE of 1.08 mm on head circumference.
- Collaborated across a 5–8 person team (researchers, clinicians, annotators); drove data collection protocols and writing for multiple 2024 publications.

Archireef, UAE 🔗 **May 2023 – August 2023**
Summer Research Intern (Computer Vision) *Abu Dhabi, UAE*

- Developed and evaluated deep learning segmentation models on spatially referenced marine imagery, achieving 60% Dice on orthomosaic test sets.
- Enhanced preprocessing pipelines under variable geographic and environmental conditions; integrated computer vision with geospatial data analytics for the ReefSFM project.

Publications 🎓

- **Qazi, Mohammad Areeb**, et al. "Continual Learning in Medical Imaging from Theory to Practice: A Survey and Practical Analysis." arXiv:2405.13482 (2024). [DOI]
- **Qazi, Mohammad Areeb**, et al. "DynaMMo: Dynamic Model Merging for Efficient Class Incremental Learning for Medical Images." arXiv:2404.14099 (2024). [DOI]
- **Qazi, Mohammad Areeb**, et al. "Multi-task Learning Approach for Unified Biometric Estimation from Fetal Ultrasound Anomaly Scans." MICCAD, Springer Singapore, 2023. [DOI]
- **Areeb, Q.M.** et al. "Filter bubbles in recommender systems: Fact or fallacy—A systematic review." Wiley WIREs Data Mining and Knowledge Discovery, 2023. [DOI]
- **Areeb, Q. M.**, Nadeem, M., Alroobaea, R., & Anwer, F. "Helping hearing-impaired in emergency situations: a deep learning-based approach." IEEE Access, 2022. [DOI]

Selected Projects

DynaMMo: Dynamic Model Merging for Incremental Learning [MIUA 2024] **[CODE]**

- Compute- and memory-efficient method to alleviate catastrophic forgetting via lightweight adapters and post-hoc model merging (see Experience above).

MedMerge: Merging Models for Effective Transfer Learning [MICCAI 2024] **[CODE]**

- Kernel-level fusion of models from diverse initializations; up to 3% F1 improvement on new medical tasks with no additional task-specific training.

XReal: Anatomy and Pathology-Aware X-ray Generation **[CODE]**

- Controllable diffusion pipeline with spatial anatomy/pathology controllers and radiologist-in-the-loop validation for synthetic chest X-ray generation.

Prompting in Audio Language Models [EMNLP 2024] **[CODE]**

- Dual-branch prompting technique (audio + language) improving few-shot performance on Audio LMs without fine-tuning.

Honors and Awards

- Won **2nd Place, UAE Hackathon 2025** — cash prize of **AED 10,000** for outstanding innovation.
- Secured **Full Scholarship** for MSc from Mohamed Bin Zayed University of Artificial Intelligence, Abu Dhabi, UAE.
- Received **Merit-Cum-Means Scholarship** for bachelor's program by the Indian government.
- **AI Lead** at Google Developer Student Club, Aligarh Muslim University, 2021.

References available on request